CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 17: Circle

Every formula and result of the chapter on one sheet

1	Diameter	$d = 2r$
2	Circumference	$2\pi r$
3	Chord relation	$r^2 = d^2 + \left(\frac{c}{2}\right)^2$
4	Length of a chord	$2\sqrt{r^2 - d^2}$
5	Distance of a chord from the centre	$\sqrt{r^2 - \left(\frac{c}{2}\right)^2}$
6	Perpendicular from the centre	Bisects the chord
7	Centre to mid-point of a chord	Perpendicular to the chord
8	Equal chords	Equidistant from the centre
9	Parallel chords, opposite sides	Distance apart $= d_1 + d_2$
10	Parallel chords, same side	Distance apart $= d_1 - d_2$
11	Common chord	Bisected at right angles by the line of centres
12	Equal arcs or chords	Subtend equal angles at the centre
13	Side of a regular n-gon	Subtends $\frac{360^\circ}{n}$ at the centre